

1. Executive Summary

This report evaluates and recommends an AI service to power the "Intelligent Research Assistant" proof-of-concept (POC). The project's goal is to discover licensed online casinos and their current *casino-specific* promotions across NJ, MI, PA, and WV, and then compare these findings against the Reel Edge database.

The project requires a hosted **AI service (API)**, not a self-hosted solution, to prioritize rapid implementation and ease of use for this POC. The service must excel at **real-time web research, extraction of structured JSON data, and simple Node.js integration.**

Based on these specific requirements, our primary recommendation is to use the **Google Gemini Pro API**. It offers the best combination of real-time, verifiable web research (via "Grounding with Google Search") and state-of-the-art reasoning. This is ideal for discovering official casino licensing information and accurately distinguishing casino from sports promotions.

2. Project Goal and AI Requirements

The selected AI service must fulfill two primary functions:

- 1. **Web Research:** Act as a researcher to find licensed casinos by state (prioritizing official sources like gaming commissions) and discover their *current* promotional offers.
- 2. **Data Extraction:** Understand the details of promotions (e.g., bonus amount, type), distinguish them from sports-related offers, and output the findings in a structured JSON format for the UI.

3. AI Model Comparison

The following table provides a high-level overview of potential models, including both the API-based services we are targeting and the self-hosted options we are excluding for this POC.

Model	Why It's a Good Match	Pros	Cons
-------	-----------------------	------	------

1. Perplexity API (API-Based)	This model is specifically designed for real-time web research and answering questions with citations. It is one of the best tools for your primary task (finding casinos and promos).	* Best-in-Class Research: Its entire purpose is to search the web in real-time and synthesize answers. * Cites Sources: Automatically tells you <i>where</i> it found the info, which is perfect for verifying a casino's license. * Easy Integration: Has a clear API and TypeScript/Node.js support. * Low Cost: Very competitive pricing (e.g., \$5 per 1,000 requests).	* NOT self-hosted. * Less of a "general purpose" model (like GPT-4) and more of a specialized research engine.
2. Google Gemini Pro (API-Based)	Natively integrated with Google Search, this model has unparalleled access to real-time web information. It's perfect for finding obscure official sources (like state gaming commissions).	* Excellent Web Access: Uses "Grounding with Google Search" for finding data and providing citations. * Great at JSON: Strong at "function calling" to return structured JSON data for your UI. * Easy to Use: Official @google/genai Node.js library.	* NOT self-hosted. * Cost is per-query (though a free tier is available for development).

3. OpenAI GPT-4o (API-Based)	An industry-standard, all-around model with excellent web browsing and world-class reasoning. It will be very reliable at both finding info and comparing promos.	* Reliable Browsing: Built-in web search tool is fast and effective. * Top-Tier Reasoning: Great at understanding ambiguous promo terms. * Easy Integration: The openai Node.js library is extremely simple to use.	* NOT self-hosted. * Can be more expensive than other options at scale.
4. Mistral Large 2 (Self-Hosted)	This is one of the most powerful open-source models you can self-host. It is known for its excellent performance and strong function-calling abilities.	* Meets "self-hosted" requirement. * Excellent JSON Output: Specifically noted for "native JSON output" and "advanced function calling capabilities", which is perfect for your UI. * No per-query costs (after setup).	* NO built-in web access. You <i>must</i> code a tool to connect it to a search API. * Very hard to implement. Requires powerful GPUs and complex setup. Not "easy."
5. Meta Llama 3.1 (70B) (Self-Hosted)	This is the other top-tier open-source model. It has incredible reasoning abilities and is very good at following instructions (like	* Meets "self-hosted" requirement. * Amazing Instruction Following: Excellent at adhering to your	* NO built-in web access. Same as Mistral, you must build the web research tool yourself. * Extremely hard to implement.

	"return only casino promos").	"no sports" rule. * Strong Function Calling: Very reliable for outputting structured JSON.	Requires massive hardware (e.g., over 48GB VRAM) and is very complex.
--	-------------------------------	--	---

4. Recommendation and Justification

As per project clarification, the goal for this POC is to use an **AI service (API), not a self-hosted model**. This requirement prioritizes speed of implementation, ease of use in Node.js, and removes the significant cost and complexity of managing GPU infrastructure.

Therefore, we can immediately disqualify **Mistral Large 2** and **Meta Llama 3.1 (70B)** from consideration for this POC. While powerful, they fail the "easy to implement" requirement and, crucially, **do not have built-in web access**. They would require us to build a separate web research tool, violating the project's "no web scrapper" principle.

This leaves the three API-based services.

- **Primary Recommendation: Google Gemini Pro (API-Based)**
This model provides the optimal balance. Its native integration with Google Search is a key advantage for finding obscure official sources like state gaming commission websites. Its strong, native JSON output capabilities align perfectly with the need to feed structured data to the UI.
- **Runner-Up: OpenAI GPT-4o**
This is a highly reliable and proven "all-rounder". Its reasoning is top-tier, and its web browsing tool is effective and easy to use. It is a very safe and effective choice.
- **Note on Perplexity API:**
While its research capabilities are excellent, it is a highly specialized tool. For the POC, using a more general-purpose model like Gemini or GPT-4o provides more flexibility for potential future tasks (e.g., more complex promo comparison logic).

5. Next Steps

We recommend proceeding with the **Google Gemini Pro API**. The immediate next step is to

validate this choice against the test cases outlined in the IA-model.md document:

1. Casino Discovery Test
2. Promotion Extraction Test
3. Simple & Complex Comparison Tests
4. Structured Output Test

Would you like me to provide the sample Node.js code for the "Casino Discovery Test" using the Google Gemini API?

This video provides a practical demonstration of how to use Mistral's models, which can be helpful for understanding the "self-hosted" options mentioned in the table.

Watsonx.ai Demo: Mistral Large 2